

SMART INDIA HACKATHON 2026



Problem Statement ID – SIH26003

Problem Statement Title – AI-Driven Cognitive Support & Dementia Care in NER

Theme – MedTech / HealthTech

PS Category – Software

Team ID – Internal Hackathon

Team Name – SAHAYAK

Team Members :

- 1. Krishraj Singh Chauhan (Team leader)**
- 2. Megha Jain**
- 3. Kartik Jain**
- 4. Nandini Bisen**
- 5. Mahima Sanodiya**
- 6. Eshika Mahore**



Idea Title & proposed solution

A simple, multilingual cognitive-care companion for patients & caregivers

PROPOSED SOLUTION

- Patient portal: voice guidance, memory games, reminders and SOS.
- Caregiver dashboard: response-time trends, adherence and alerts.

HOW IT ADDRESSES THE PROBLEM

- Reduces daily confusion with familiar prompts and routines.
- Gives caregivers objective trend signals instead of guesswork.
- Designed for low-resource, mobile-first use in NER communities.

INNOVATION & UNIQUENESS

- Regional anchoring: Assam tea, rhino and local-language support.
- Latency tracking measures task-response trends over time — not a diagnosis.
- One-tap SOS + location sharing connects care with safety.

Patient → SMRITI-NER → Caregiver | Support • Monitor • Respond

TECHNOLOGY STACK

Frontend : HTML5 • Tailwind CSS • JavaScript

Voice : Web Speech API / SpeechSynthesisUtterance

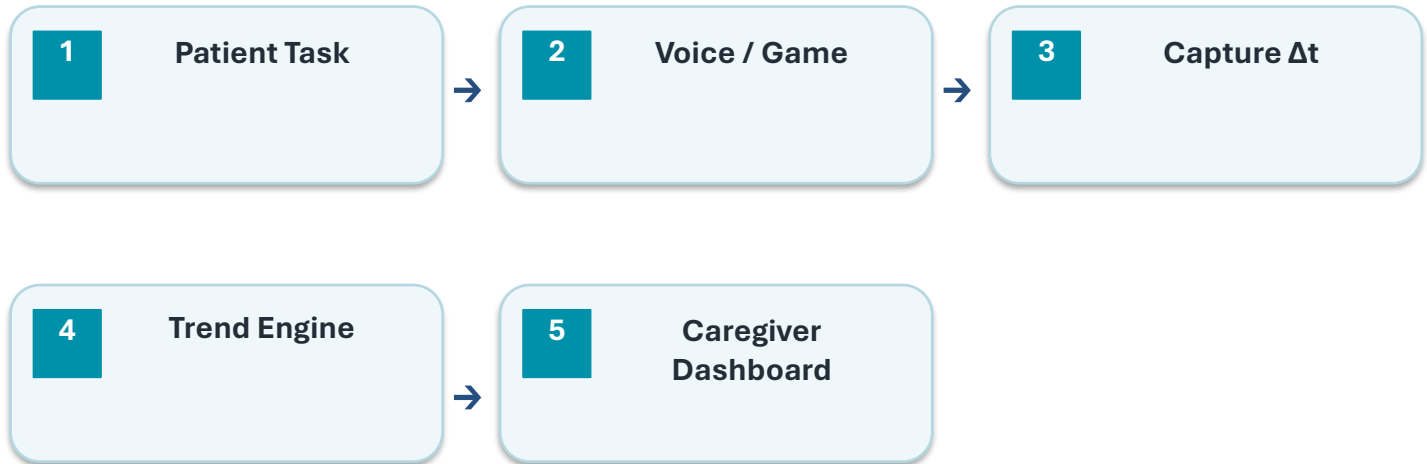
Telemetry : Chart.js • response-time logging

Data scale: Firebase (planned) • secure caregiver data

Deployment : GitHub + Netlify CI/CD

Regional language : English, Hindi, Assamese

METHODOLOGY / SYSTEM FLOW



Live demo <https://smritiner.netlify.app/>

Netlify link

Source code <https://github.com/cartickkk/smriti-ner.git>

GitHub Repository link



FEASIBILITY AND VIABILITY



FEASIBILITY

Low-cost deployment

Browser-first software; no dedicated hardware required. Works on basic smartphones.

Offline-ready

Cache core games, prompts and reminders; sync telemetry when connectivity returns.

Accessible by design

Large touch targets, simple navigation, voice guidance and high-contrast UI.

CHALLENGES & RISKS → MITIGATION

Connectivity	Offline cache + delayed sync	Limited network in remote areas
Language coverage	Bhashini integration roadmap + fallback Hindi/English	Regional dialect diversity
Privacy & safety	Consent, minimum data collection, secure auth; SOS confirmation	Sensitive caregiver data
Clinical interpretation	Present trends as support signals; clinician validation before diagnostic use	False alarms / over-interpretation

TARGET IMPACT

PATIENTS

Greater independence in daily routines

Voice prompts • memory games • medicine reminders • SOS

CAREGIVERS

Earlier visibility into changing patterns

Trend charts • adherence view • emergency alerts

NER COMMUNITIES

Affordable, culturally relevant digital support

Mobile-first • multilingual pathway • rural-ready

EXPECTED VALUE CHAIN

Simple routines

↓ confusion



Repeated tasks

response trends



Caregiver insight

faster response



Safer daily living

better support

Designed as a care-support tool; clinical decisions remain with qualified professionals.

RESEARCH BASE

WHO — Dementia<https://www.who.int/health-topics/dementia>

Problem context, caregiver burden and dementia-care priorities.

W3C — WCAG 2.1<https://www.w3.org/TR/WCAG21/>

cognitive, physical and mAccessibility guidance form mobile users.

Web Speech API — MDNhttps://developer.mozilla.org/en-US/docs/Web/API/Web_Speech_API

Browser speech synthesis/recognition implementation reference.

Bhashini — Government of India<https://bhashini.gov.in/>

Planned pathway for Indian-language speech/translation integration.

MoCA — Montreal Cognitive Assessment<https://mocacognition.com/>

Reference for established cognitive-assessment concepts; SMRITI-NER is not a diagnostic test.